Sign-Language HUD Glasses

Hardware bring-up guide: test every part before building.

Branch `testout` · 2026-10-24 · Charan · Pictures are drawn diagrams, not photos.

How to use this guide

1. Do the steps in order: 1 → 9.
2. Do one small action at a time.
3. Check the green **Done when** box before moving on.
4. If something fails, read the orange **If it fails** box.
5. Write every result in the checklist on the last page.

> ⚠ **Safety: read first**
>
> - **No LiPo battery** anywhere during these tests. The batteries haven't arrived, and the TP4056 needs a resistor change first (Step 7).
> - Meter red lead for volts goes in **Input HI (VΩ)**, never the **10A** jack.
> - Don't let the two meter probes touch each other while a board is powered.
> - Soldering iron (later, Step 8) is ~330 °C. Always put it back in its stand.

Parts

In hand ✅

- ESP32-C3 SuperMini ×2
- 0.96" OLED SSD1306 ×2 (4-pin I2C)
- TP4056 Type-C charger ×2
- Slide switch ×3
- Veroboard 6×4 ×2
- Male pin strip 40×2, female 40×1
- Type-C cable
- MB102 breadboard
- Jumper wires
- Soldering kit + multimeter

On the way 🚚

- LiPo 3.7 V 500 mAh ×2
- 25 mm lens (f 45 mm)
- 1 mm acrylic sheets
- Mirror piece (local shop)

Stretch (later)

- ESP32-CAM + programmer
- MT3608 boost

Where the files are

What	Path
Test sketches	`firmware/tests/t1…t4`
Send text from Mac	`tools/send_test.py`
Short guide	`docs/hardware_bringup.md`
Daily log	`docs/handoff/YYYY-MM-DD.md`

Step 1: Check the boards with the meter ✅ DONE

Goal: make sure both ESP32-C3 boards give the right voltages.

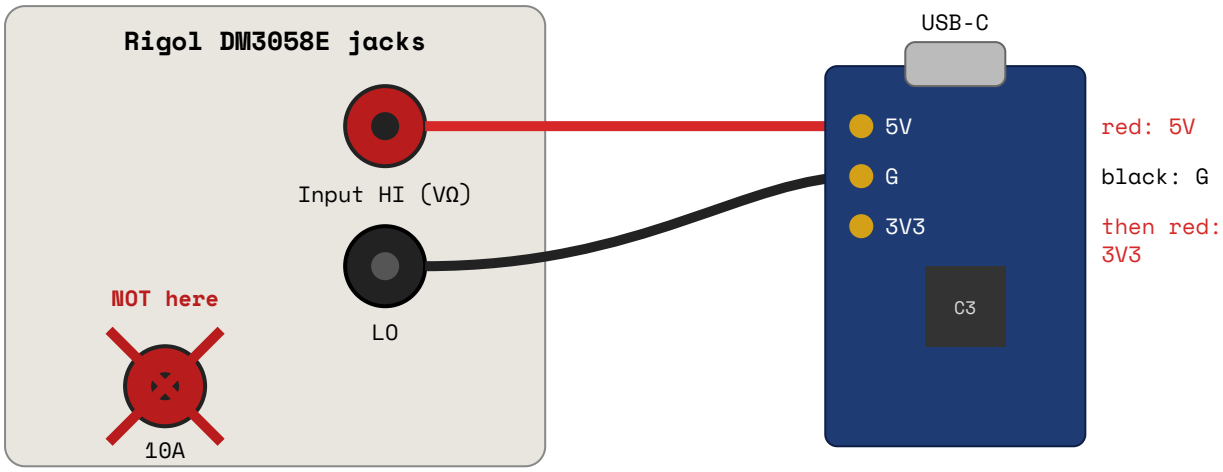


Diagram: red lead in Input HI, black in LO, the 10A jack is never used for volts.

1. Red lead into **Input HI (VΩ)**. Black lead into **LO**.
2. Press the ⎓**V** button (DC volts). It turns green.
3. Plug the ESP32-C3 into the Mac with the Type-C cable.
4. Black probe on the **G** pad.
5. Red probe on the **5V** pad. Read the number.
6. Keep black on G. Move red to **3V3**. Read the number.
7. Repeat for board #2.

Done when
5V reads 4.8–5.2 V and 3V3 reads 3.2–3.4 V on both boards.
Your result: 5V = **5.162 V**, 3V3 = **3.302 V** on both boards ✅

If it fails
Reads about 0 V → the red lead is in the 10A jack; move it to Input HI. No reading → try another USB port or cable.

Step 2: Put the ESP32-C3 on the breadboard (no solder)

Goal: the board sits on pins in the breadboard without soldering. Soldering comes at the very end (Step 8).

2a. Snap the pins

DONE

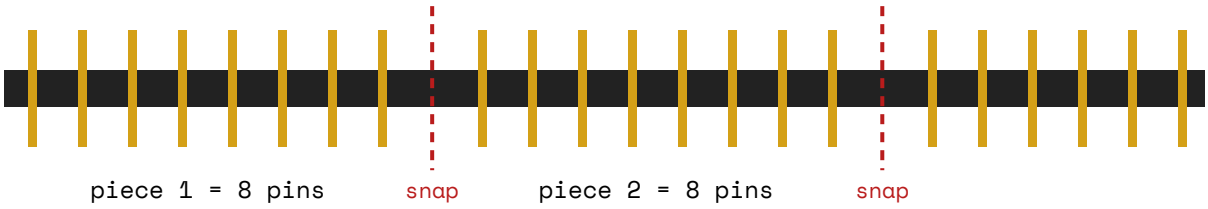


Diagram: snap the strip after every 8th pin to get two pieces.

2b. Pins into the board

1. Hold the ESP32-C3 with the chip side up.
2. From underneath, push the **short end** of piece 1 into one row of holes.
3. Push piece 2 into the other row. The long pins point down.

2c. Board onto the breadboard

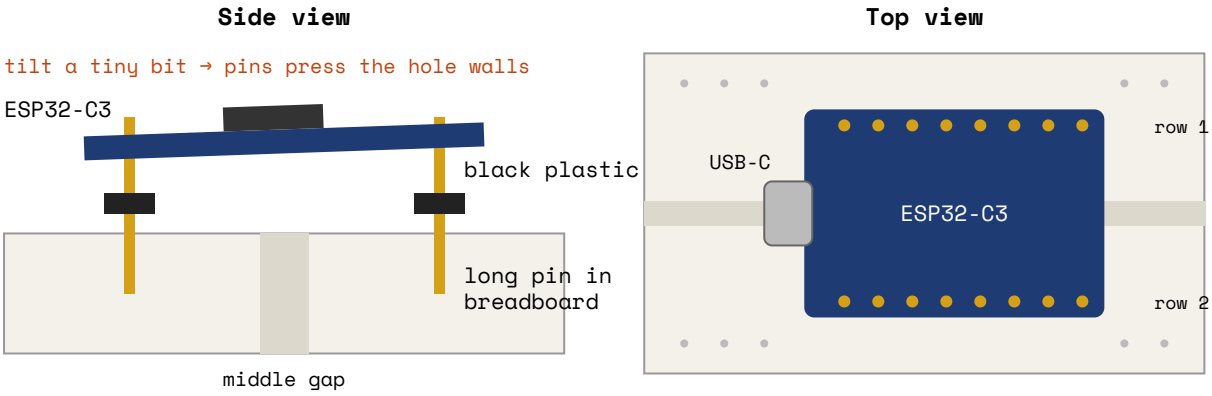


Diagram: one pin row on each side of the middle gap, USB-C toward the breadboard edge.

1. Turn the board so the **USB-C port** points to the edge of the breadboard.
2. Line it up across the **middle gap**: one pin row on each side.
3. Thumbs on the board edges, not on the chip. Press straight down.
4. Tilt it a tiny bit so the pins press against the hole walls.

2d. Check the OLED pins

1. Look at the OLED's 4 pins: **VCC GND SCL SDA** (order may differ, read the labels).
2. Already soldered → plugs straight into the breadboard.
3. Loose pins → same trick as 2b.

Done when
The board sits flat, doesn't wobble, and Step 3 (blink) works with it on the breadboard.

If it fails (later steps)
Random resets, upload errors or "no I2C devices" → **press the board down first**, before blaming the code.

Step 3: Is the ESP32-C3 alive? (blink)

Goal: upload the first program and see the LED blink. Sketch: `firmware/tests/t1_blink`

3a. One-time setup

1. Install **Arduino IDE 2** from arduino.cc.
2. Open it. Left bar → **Boards Manager**.
3. Search **esp32**. Install **esp32 by Espressif Systems**.

3b. Settings

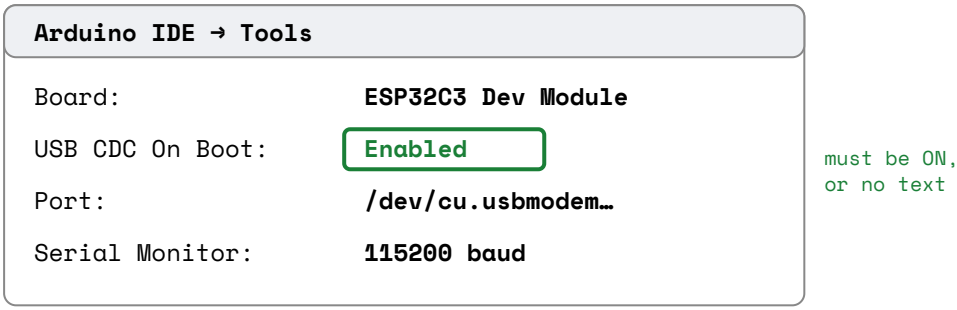


Diagram: the four settings that matter.

1. Plug in the board (USB-C).
2. Set the four settings above.

3c. Upload

1. File → Open → `firmware/tests/t1_blink/t1_blink.ino`.
2. Click **Upload** (the → arrow).
3. Wait for **Done uploading**.
4. Open **Serial Monitor**, set 115200.



Diagram: what you should see.

Done when
The blue LED blinks once per second and Serial Monitor prints `alive 0`, `alive 1`, `…`. Do it for both boards.

If it fails
Upload error → hold **BOOT**, tap **RST**, let go of BOOT, upload again.
No port → try another cable (some only charge).
No text → USB CDC On Boot is not Enabled.
Random disconnects → press the board down (friction fit).

Step 4: Connect the OLED screen

Goal: the ESP32-C3 finds the OLED, then shows text on it.

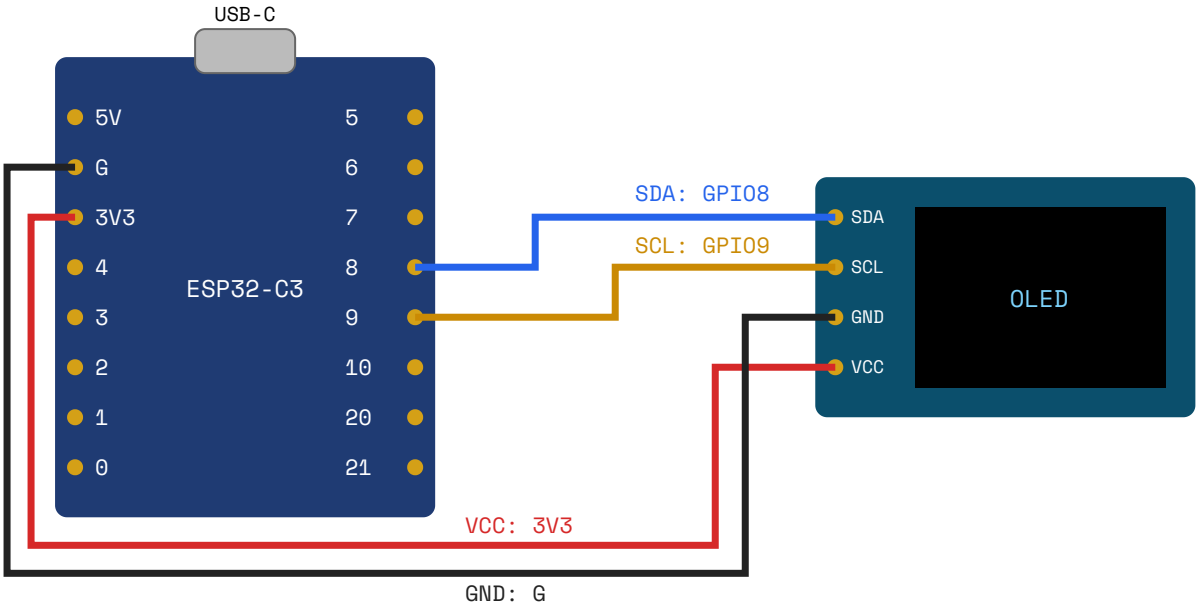


Diagram: red VCC→3V3, black GND→G, blue SDA→GPIO8, yellow SCL→GPIO9. Check the pin labels printed on your own boards.

OLED pin	ESP32-C3 pin	Wire colour
VCC	3V3 (not 5V)	red
GND	G	black
SDA	GPIO8	blue
SCL	GPIO9	yellow

4a. Wire it

1. Unplug USB first.
2. Push the OLED into the breadboard, away from the ESP32-C3.
3. Add the 4 jumper wires from the table. Use the free holes in the same column as each pin.
4. Check every wire twice. Then plug USB back in.

4b. Find the OLED (`t2_i2c_scan`)

1. Upload `firmware/tests/t2_i2c_scan`.
2. Open Serial Monitor.

```
found device at 0x3C
---
```

4c. Show text (`t3_oled_hello`)

1. Library Manager → install **Adafruit SSD1306** (say yes to Adafruit GFX).
2. Upload `firmware/tests/t3_oled_hello`.
3. Mirror test: set `#define MIRROR_X 1`, upload, look at the OLED in a mirror.
4. Swap in OLED #2 and repeat 4b + 4c.



Diagram: normal screen (left), mirrored screen (right).

Done when
Scan shows `0x3C`, the screen shows **HELLO** + a counter, and mirrored text reads correctly in a mirror. Both OLEDs.

If it fails
"no I2C devices" → SDA/SCL swapped, loose wire, or VCC on 5V. Press the board down.
Scan shows `0x3D` → change `OLED_ADDR` to `0x3D` in t3 and t4.
Screen blank but scan OK → Adafruit SSD1306 library missing or wrong address.

Step 5: Connect the ESP32-C3 to Wi-Fi

Goal: the glasses board joins your phone hotspot and shows its IP address. Sketch: `firmware/tests/t4_wifi_udp`

5a. Phone hotspot

1. Turn on your phone hotspot.
2. iPhone: Settings → Personal Hotspot → **Maximize Compatibility ON** (2.4 GHz).
3. Android: hotspot band = **2.4 GHz**.

5b. Wi-Fi password file

1. Open folder `firmware/tests/t4_wifi_udp/`.
2. Copy `secrets.h.example` → name the copy `secrets.h`.
3. Type your hotspot name and password inside.
4. `secrets.h` is never committed (gitignored). Keep it that way.

```
#define WIFI_SSID "your-hotspot-name"
#define WIFI_PASS "your-hotspot-password"
```

5c. Upload

1. Upload `t4_wifi_udp`.
2. Watch the OLED.

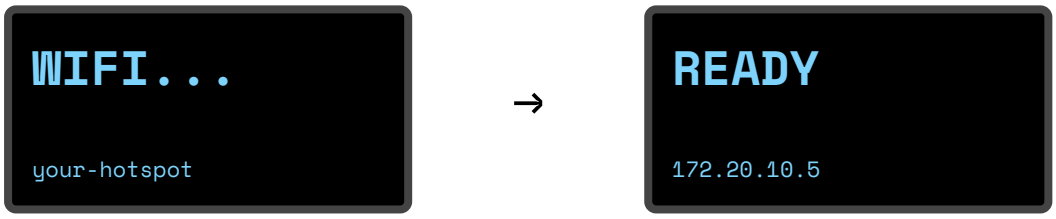


Diagram: connecting (left), connected with its IP address (right). Your IP will differ.

3. Write down the IP shown on the OLED.

Done when
The OLED shows **READY** and an IP address.

If it fails
Stuck on **WIFI...** → wrong password, hotspot is 5 GHz, or the board is too far from the phone. The SuperMini antenna is weak; keep the phone within 1–2 m.

Step 6: Send a word from the Mac to the OLED

Goal: a word typed on the laptop shows up on the glasses screen. This is the same link the sign recogniser will use.

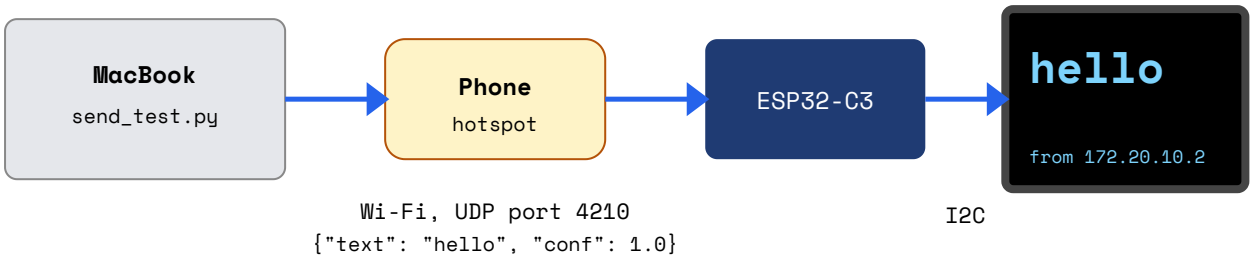


Diagram: the word travels Mac → hotspot → ESP32-C3 → OLED.

1. Connect the Mac to the **same** phone hotspot.
2. Open Terminal in the project folder.
3. Send one word (use the IP from Step 5):

```
venv/bin/python tools/send_test.py --host 172.20.10.5 --text hello
```

4. Look at the OLED.
5. Try the name instead of the IP:

```
venv/bin/python tools/send_test.py --text hello
```

6. Try typing many words (Ctrl+C to stop):

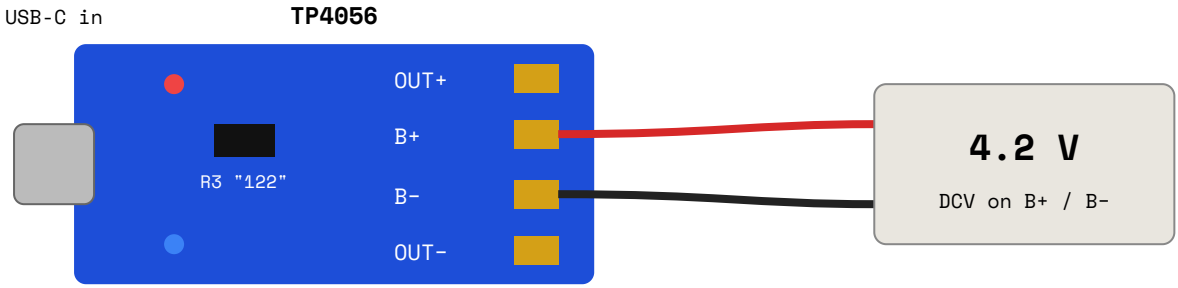
```
venv/bin/python tools/send_test.py --host 172.20.10.5
```

Done when
hello appears on the OLED in under a second.

If it fails
Nothing appears → Mac is on a different Wi-Fi; check the IP again.
IP works but `glasses.local` doesn't → fine for now, use the IP and note it in the daily log.

Step 7: Test the chargers and switches (USB only)

Goal: both TP4056 chargers and all 3 slide switches work. **No battery.**



R3 sets charge current: 1.2 kΩ = 1 A (too high for 500 mAh)

Diagram: measure B+ (red) and B- (black) with only USB plugged in.

7a. Chargers

1. Plug USB-C into TP4056 #1. Nothing else connected.
2. LEDs may flicker or both glow. That's normal with no battery.
3. Meter on DC volts. Red on **B+**, black on **B-**. Read.
4. Red on **OUT+**, black on **OUT-**. Read (0 V is OK without a battery).
5. Repeat for TP4056 #2.

7b. Slide switches

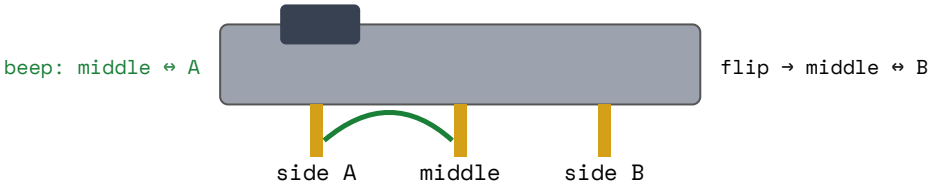


Diagram: the middle pin connects to one side, flip it and it connects to the other.

1. Meter on beep mode ·))).
2. Probes on **middle** pin and **side A** → beep (or not).
3. Flip the switch. Now **middle** and **side B** should beep.
4. Test all 3 switches.

Done when
B+ reads 4.1–4.25 V on both chargers, and every switch swaps sides when flipped.

⚠ **Before the LiPo batteries arrive**
Resistor **R3** (marked `122` = 1.2 kΩ) makes the TP4056 charge at **1 A**. That's too much for a 500 mAh LiPo. Change it to **~2.4 kΩ (≈ 500 mA)**: remove the `122` part and put a 2.2–2.4 kΩ resistor between the PROG pad and GND. Measure it in Ω mode before use. Ask for help with this step; it's fiddly.

Step 8: Final build: soldering LATER

Goal: permanent, clean pins. Do this only after every test above passes and all parts have arrived.

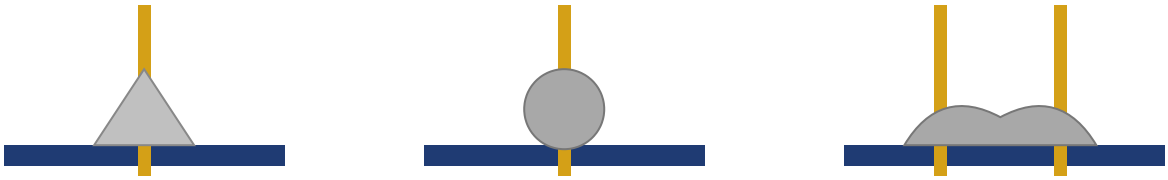


Diagram: a good joint, a ball (not enough heat), and a bridge (short circuit).

1. Pins in the breadboard, board on top (same as Step 2). This keeps them straight.
2. Heat the iron for 2 minutes (~330 °C). Wipe the tip on the wet sponge.
3. Touch the iron to the **pin and the copper ring together**. Count 1 second.
4. Push solder onto the joint (not the iron) until a small cone forms.
5. Remove the solder first, then the iron.
6. Do one corner pin, check the board is flat, then the rest.
7. Unplug the iron. Let the board cool for 1 minute.
8. Beep mode: test every pair of neighbouring pins → **no beep**.
9. Run Steps 3 and 4 again.

Done when
All joints are shiny cones, no neighbouring pins beep, and blink + OLED work again.

If it fails
Bridge → add flux, drag the clean hot tip between the pins. Ball → reheat with a touch of flux.

Step 9: Results checklist

Tick each box as you go. Copy the results into `docs/hardware_bringup.md` and today's daily log.

#	Part	Test	OK	Notes / reading
1	ESP32-C3 #1	Volts 5V / 3V3	☑	5.162 V / 3.302 V
2	ESP32-C3 #2	Volts 5V / 3V3	☑	5.162 V / 3.302 V
3	Pin strips	2 × 8 pins snapped	☑	
4	ESP32-C3 #1	Friction fit on breadboard	☐	
5	ESP32-C3 #1	Blink + serial (Step 3)	☐	
6	ESP32-C3 #2	Blink + serial (Step 3)	☐	
7	Type-C cable	Upload works (data)	☐	
8	OLED #1	Scan 0x3C, HELLO, mirror	☐	
9	OLED #2	Scan 0x3C, HELLO	☐	
10	Wi-Fi	READY + IP on OLED	☐	IP:
11	UDP link	send_test.py → OLED	☐	glasses.local: yes / no
12	TP4056 #1	B+ ≈ 4.2 V	☐	
13	TP4056 #2	B+ ≈ 4.2 V	☐	
14	Slide switches ×3	Middle ↔ side swaps	☐	
15	Breadboard + jumpers	No faults in Steps 4–6	☐	
16	TP4056 R3 mod	~2.4 kΩ before LiPo	☐	measured Ω:
17	Soldering	Step 8 (final build)	☐	

What comes after

1. LiPo + lens + acrylic arrive → battery power, build the housing.
2. Full glasses firmware: word-wrap, history, "NO LINK" screen.
3. Connect the sign-recognition app to the same UDP link.